

AS

CONTACT



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EDUCATION

Ph.D. | Environmental Sciences
(Soil & Water)

UC Riverside

2018 – 2021

MS | Plant Science

California State University | Fresno

2015– 2017

BSc | Agriculture

Punjab Agricultural University,
Ludhiana | Punjab, India

2011 – 2015

RELEVANT COURSES

ENSC 175 – Spatial Analysis
and Remote Sensing, CS235 –
Data mining methods,
STAT208 – Statistical Data
mining, ENSC 217 - Vadose
Zone Processes, ENSC110 –
Environmental Statistics,
PLANT 251 – Soil Plant Water
relations.

AMNINDER SINGH

Post-Doctoral Scholar

PROFESSIONAL EXPERIENCE

POST-DOCTORAL SCHOLAR

University of California, Riverside & [USDA Salinity Lab](#) | Jan 2022 – Present

Focusing on estimating soil moisture in agricultural fields in the contiguous United States by combining public and private soil moisture ground measurements, remote sensing, and other geospatial datasets.

- Use satellite data from multiple sources, including synthetic aperture radar (SAR) and multispectral imagery for agricultural applications.
- Responsible for gathering large spatial and time-series data using google earth engine's python API and implementing machine learning models for soil moisture estimation using python libraries scikit-learn, TensorFlow, etc.
- Experience working with Planet Labs Data and Orders API.
- Collaborating with private companies and data scientists to implement new strategies and fulfill project requirements.

GRADUATE STUDENT RESEARCHER

University of California, Riverside | Jan 2018 – Dec 2021

Dissertation: Advancing Urban Landscape Irrigation Management using Smart Controllers and Machine Learning-Based Models

- Experienced in sampling and analyzing soil hydrology field measurements, including soil water potential, soil volumetric water content, infiltration, evapotranspiration, and weather data.
- Developed machine learning and deep learning models using soil hydrological and time-series weather datasets.
- Collaborated and participated in several field campaigns supporting remote sensing projects.
- Gained experience with aggregating & statistically analyzing large agricultural datasets using Python, R, or MATLAB.

CO-PRESIDENT, MINI-GSA Environmental Sciences | UCR | 2020 – 2021

TEACHING ASSISTANT, Introduction to Environmental Science: Natural Resources (ENSC001), Dept. of Environmental Sciences | UCR | Fall 2020

GRADUATE RESEARCH ASSISTANT

CSU, Fresno | Aug 2015 – Dec 2018

Thesis: Use of EM-38 soil surveys in forage fields at a saline drainage water reuse site to calibrate a hydro-salinity model for decision support.

- Conducted geophysical measurements for soil salinity mapping using electromagnetic sensing platforms.
- Analysis of the data using tools like ArcGIS and R.

AMNINDER SINGH

SKILLS

- Statistical and programming skills: Python, Matlab, and R
- Experience with Python libraries: TensorFlow, scikit-learn, pandas, NumPy, matplotlib, seaborn, statsmodels, xarray, geopandas, PySpark etc.
- Deployment – Django, Streamlit.
- Machine Learning/Deep Learning
- ArcGIS, QGIS, Google Earth Engine
- Soil and environmental sensor network data acquisition systems.
- Geophysical soil sensing.
- SQL, Git.

LANGUAGES

- English
- Punjabi
- Hindi

AFFILIATIONS

- Soil Science Society of America (SSSA)
- American Society of Agronomy (ASA).
- Crop Science Society of America (CSSA).
- American Society of Agricultural and Biological Engineers (ASABE).

PROFESSIONAL EXPERIENCE (Cont...)

GRADUATE TEACHING ASSISTANT, SWL100- (Soils Lab) | CSU, Fresno | Fall 2015 & Fall 2016

ACHIEVEMENTS AND AWARDS

- Best Oral Presentation Award - 2021 UCR Environmental Sciences Graduate Student Symposium.
- Dissertation Year Program Fellowship, Department of Environmental Sciences, UCR (Fall 2021).
- Hilda and George Liebig Environmental Sciences Summer Fellowship, Department of Environmental Sciences, UCR (2020).
- SCSC-NWRI Fellowship, Southern California Salinity Coalition (SCSC) and the National Water Research Institute (NWRI), 2018-2020.
- Stolzy-Letey Travel Scholarship, Department of Environmental Sciences, UCR (2018-2019).
- Mini GSA Award, Hilda and George Liebig Foundation, Department of Environmental Sciences, UCR (2019, 2021).
- UCANR Drone Camp, June 18-20, 2018, UC San Diego
- Outstanding Thesis Award. Jordan College of Agricultural Sciences & Technology, CSU, Fresno (2017-2018 academic year).
- Graduate Student Achiever, Department of Plant Science, CSU, Fresno. (2018).
- Ag One-John P. "Phil" Larson Scholarship, CSU, Fresno (Spring 2017).
- Inducted to The Honor Society of Phi Kappa Phi (Spring 2017).
- Jim Pattersen's Ag Science student recognition award - Agriculture Science Student of the Year, 23rd Assembly District, (2017).
- The Gerald O. Mott Meritorious Graduate Student Award in Crop Science, Crop Science Society of America (CSSA), (2017).
- Third place in Poster competition, Plant and soil conference – American Society of Agronomy, California chapter, (2017).
- Jordan Assistantship, Jordan College of Agricultural Sciences and Technology, CSU, Fresno (August 2015-May 2017).
- Graduate Research Fellowship, Graduate Net Initiative, CSU, Fresno (2016-2017).
- Travel grant, Division of Graduate Studies, CSU, Fresno for the 2016 ASA, CSSA, and SSSA Annual Meeting, November 5-9, 2016, in Phoenix, AZ.
- Dean's Scholarship Tuition Waiver, CSU, Fresno (2015- 2016).
- Punjab State Marketing Board Scholarship, PAU, Ludhiana (2011 - 2015).

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PRESENTATIONS

- **Singh A., Haghverdi A., Sapkota A., Haver D.,** Recycled Water Application for Turfgrass Irrigation to conserve water and maintain turf health, **ASABE 2021 Annual International Meeting** | July 12–16, 2021 | Virtual (Poster presentation).
- **Singh A., Haghverdi A., Sapkota A., Haver D.,** Autonomous Irrigation in Bermudagrass Using Recycled Water and Soil Moisture Sensor Based Smart Controller, **2020 ASA-CSSA-SSSA International Annual Meeting** | November | Virtual (Oral presentation)
- **Singh A., Haghverdi A. Öztürk H. S., Durner W.,** Pseudo Continuous Water Retention Pedotransfer Functions for International Soils Measured with the Evaporation Method and the HYPROP System, **2020 ASA-CSSA-SSSA International Annual Meeting** | November | Virtual (Poster)
- **Singh A., Haghverdi A., Sapkota A., Haver D.,** Autonomous Bermudagrass irrigation using recycled water and SMS based smart controller, **UC Riverside Urban landscape Irrigation and Water Conservation Virtual Field Day for Master Gardeners** | September 3, 2020.
- **Singh A., Sharifi M., Ghodsi S., Haghverdi A.,** Monitoring the impact of multiple water conservation strategies on turfgrass using ground based remote sensing tools. **2019 SSSA Annual Meeting, San Diego, CA** (Oral presentation).
- **Benes S.E., Singh A., Quinn N.W. and Cassel Sharma F.** (2018). Monitoring Soil Salinity using EM-38 Surveys to Calibrate a Hydro-salinity Model for Decision Support. 2018, The Future of Water for Irrigation in California and Israel Workshop. Davis, California, July 16-18, 2018. (Poster)
- **Singh, A., Benes, S.E., Quinn, N., Cassel, F.** Use of EM-38 Soil Surveys in Forage Fields at a Saline Drainage Water Reuse Site to Calibrate a Hydro-salinity Model for Decision Support. **Plant and Soil Conference by American Society of Agronomy – California Chapter** (2018) (Poster).
- **Singh, A., Benes, S.E., Quinn, N., Cassel, F.** Monitoring Soil Salinity in Alfalfa and ‘Jose’ tall wheatgrass fields using EM-38 soil Surveys and Developing Input Data for a Transient Hydro-salinity Computer Model. **38th Annual Central California Research Symposium**, University Business Center, CSU, Fresno, April 18, 2017 (Oral presentation).
- **Singh, A., Benes, S.E., Quinn, N., Cassel, F.** Use of EM-38 soil salinity surveys to develop validation data sets for a transient hydro-salinity model CSUID-II. **CWEMF annual meeting**, Folsom, CA, March 20-22, 2017 (Poster).
- **Singh, A., Benes, S.E., Quinn, N., Cassel, F.** Development of validation data sets for a transient hydro-salinity model using EM-38 soil surveys, irrigation water monitoring and forage analysis. **Plant and Soil Conference by American Society of Agronomy – California Chapter**, Jan. 31-Feb. 1, 2017 (Poster).
- **Singh, A., Benes, S.E., Quinn, N., Cassel, F., Bottino, U. Jr.,** Soil Salinity Mapping Using Electromagnetic Induction (EM-38) to Provide Input Data for the CSUID-II Model, a Decision Support Tool for Irrigation in Saline Water Reuse Areas. **2016 ASA, CSSA, and SSSA Annual Meeting**, Phoenix, AZ, November 6-9, 2016 (Poster).

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PUBLICATIONS (peer-reviewed)

- Haghverdi, A., M. Reiter, **A. Singh**, and A. Sapkota. 2021. Hybrid Bermudagrass and Tall Fescue Turfgrass Irrigation in Central California: II. Assessment of NDVI, CWSI, and Canopy Temperature Dynamics. *Agronomy* 11(9): 1733 Available at <https://www.mdpi.com/2073-4395/11/9/1733>.
- Haghverdi, A., M. Reiter, A. Sapkota, and **A. Singh**. 2021. Hybrid Bermudagrass and Tall Fescue Turfgrass Irrigation in Central California: I. Assessment of Visual Quality, Soil Moisture and Performance of an ET-Based Smart Controller. *Agronomy* 11(8): 1666.
- **Singh, A.**, A. Haghverdi, H.S. Öztürk, and W. Durner. 2021. Developing Pseudo Continuous Pedotransfer Functions for International Soils Measured with the Evaporation Method and the HYPROP System: II. The Soil Hydraulic Conductivity Curve. *Water* 13(6): 878 Available at <https://www.mdpi.com/2073-4441/13/6/878>.
- Haghverdi, A., **A. Singh**, A. Sapkota, M. Reiter, and S. Ghodsi. 2021. Developing irrigation water conservation strategies for hybrid bermudagrass using an evapotranspiration-based smart irrigation controller in inland southern California. *Agric. Water Manag.* 245(April 2020): 106586 Available at <https://doi.org/10.1016/j.agwat.2020.106586>.
- **Singh, A.**, A. Haghverdi, H.S. Öztürk, and W. Durner. 2020. Developing pseudo continuous pedotransfer functions for international soils measured with the evaporation method and the hyprop system: I. the soil water retention curve. *Water (Switzerland)* 12(12): 1–17 Available at <https://www.mdpi.com/2073-4441/12/12/3425>.
- **Singh, A.**, N.W.T. Quinn, S.E. Benes, and F. Cassel. 2020. Policy-Driven Sustainable Saline Drainage Disposal and Forage Production in the Western San Joaquin Valley of California. *Sustain.* 2020, Vol. 12, Page 6362 *12*(16): 6362 Available at <https://www.mdpi.com/2071-1050/12/16/6362/htm>
- **Singh, A.**, A. Haghverdi, M. Nemati, and J. Hartin. 2020. Efficient Urban Water Management Smart Weather-Based Irrigation Controllers. *Univ. Calif. Agric. Nat. Resour.* 8674(July): 1–11